

UNIVERSITI KEBANGSAAN MALAYSIA
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Short-Channel Effects in Nanoscale CMOS

KKKC6353 Nanoelectronic Devices — Lecture 1, Document 3 of 3

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The argument of this document

Short-channel effects are usually taught as a list. Five of them, or nine, depending on the textbook: channel-length modulation, DIBL, punchthrough, subthreshold leakage, surface scattering, velocity saturation, impact ionisation, hot carriers, oxide breakdown. Learn the list, reproduce the list.

That framing is wrong, and it is why the topic feels like bookkeeping.

There is one problem. The gate is supposed to be the only electrode that controls the potential barrier between source and drain. When the channel is long, it is. When the channel is short, the drain — which is a conductor held at a voltage, sitting a few nanometres away — begins to control that barrier too. Everything on the list above is a symptom of the gate losing its monopoly, or a consequence of the high fields that come with squeezing a volt into a few tens of nanometres.

And because there is one problem, there is one family of solutions. Section 6 shows that halo implants, silicon-on-insulator, FinFETs and gate-all-around nanosheets are all the same move: reduce the volume of semiconductor the drain can reach into, so the gate wins. **The rest of the module starts where that move runs out.** This document is where it gets its motivation.

Notation follows Document 2 without exception. $U_T = kT/q$ is the thermal voltage; V_T is the threshold voltage; $\mathcal{E}$ is electric field; L is *physical* gate length, not a node name.

1 One problem: electrostatic integrity

1.1 When is a channel "short"?

The textbook answer is that a device is short-channel when L becomes comparable to the source and drain depletion widths. That is a symptom, not a criterion, and it gives no design guidance.

The useful criterion comes from solving Poisson's equation in two dimensions in the channel region. Doing so produces a **natural length** λ — also called the scale length — which is the characteristic distance over which the drain's field penetrates into the channel. For a single-gate device with body thickness (or depletion depth) t_{Si} and dielectric thickness t_{ox}:

$$\lambda = \sqrt{\frac{\varepsilon_{Si}}{\varepsilon_{ox}}\, t_{Si}\, t_{ox}}$$

The potential perturbation caused by the drain decays roughly as $\exp(-L/2\lambda)$ along the channel. The design rule that follows is simple and is used throughout the industry:

$$\boxed{L \gtrsim 5\lambda \text{ to } 10\lambda}$$

Below that, the drain is no longer electrostatically far away, and the device degrades in all the ways catalogued in Section 2.

Two features of the formula do the work. First, λ falls when the body gets **thinner** — less silicon for the drain field to reach through. Second, λ falls when the dielectric gets **electrically thinner** — stronger gate coupling. Both are square-root dependences, so both are weak levers, which is why the industry eventually had to change the *geometry* rather than keep tuning the numbers.

Adding gates changes the prefactor. For a double-gate device the drain field is screened from two sides, and for a gate-all-around device from four:

$$\lambda_{\text{single}} = \sqrt{\frac{\varepsilon_{Si} t_{Si} t_{ox}}{\varepsilon_{ox}}}, \qquad \lambda_{\text{double}} = \sqrt{\frac{\varepsilon_{Si} t_{Si} t_{ox}}{2\varepsilon_{ox}}}, \qquad \lambda_{\text{GAA}} \approx \sqrt{\frac{\varepsilon_{Si} t_{Si} t_{ox}}{4\varepsilon_{ox}}}$$

Put numbers in. With EOT = 1 nm:

Architecture	Body	λ	$L_{min} \approx 5\lambda$
Planar bulk	$x_{dm} = 20$ nm	7.8 nm	≈ 39 nm
FinFET / double gate	$t_{Si} = 6$ nm	3.0 nm	≈ 15 nm
GAA nanosheet	$t_{Si} = 5$ nm	1.9 nm	≈ 10 nm

Compare with what happened. Planar bulk was pushed to physical gate lengths of about 35 nm (Intel's 65 nm process used a 35 nm gate in 2005) and then squeezed further only with aggressive halo doping and very thin dielectrics, before the industry abandoned it in the early 2010s. FinFETs then carried logic from the 22 nm class to the 3 nm class. And for gate-all-around devices, the 2022 International Roadmap for Devices and Systems projects that physical gate length will **saturate at around** 12 **nm "due to worsening electrostatics"** — which is the last row of this table.

The body thicknesses above are physically reasonable choices, not fits, but they *are* choices, and the 5λ rule is a rule of thumb. What the equation gets right is the ordering of the transitions and the scale at which each architecture runs out. That is a great deal to get from one line, and it is the most useful idea in this document.

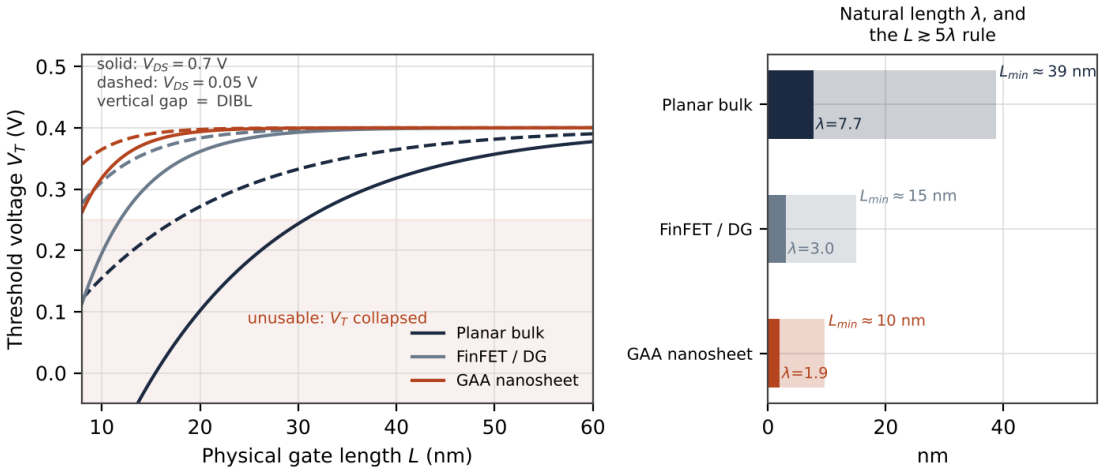


Figure 3.1. *Left: threshold voltage against physical gate length for the three architectures, computed from* $V_T(L) = V_{T\infty} - (a + bV_{DS})\exp(-L/2\lambda)$ *with identical a and b and only* λ *changed. The vertical gap between the solid and dashed curves is DIBL. Right: the natural length of each architecture and the resulting minimum usable gate length.*

1.2 The classical picture, and why it is only a heuristic

Before scale-length theory, short-channel threshold shift was explained by **charge sharing**. The long-channel V_T derivation in Document 2 assumed that *all* the depletion charge under the gate is supported by the gate. In a short device it is not: the source and drain junction depletion regions already support some of it, so the gate has less work to do, so V_T falls.

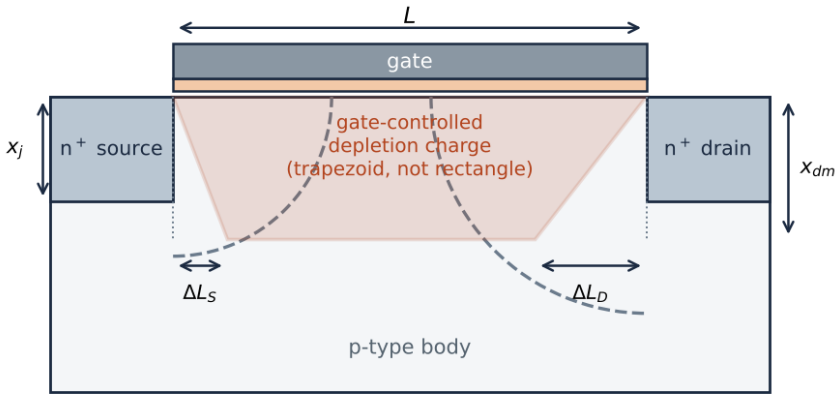


Figure 3.2. *Charge sharing in a short-channel MOSFET. The gate-controlled depletion charge is a trapezoid rather than a rectangle, because the source and drain junction depletion regions (dashed) already support the charge in the corner regions.*

Geometrically, with junction depth x_j and depletion depths x_{dS}, x_{dD}:

$$\Delta L_{S,D} \approx x_j \left(\sqrt{1 + \frac{2x_{dS,dD}}{x_j}} - 1 \right),$$

and the threshold shift is

$$\Delta V_T = -\frac{\sqrt{2q\varepsilon_{Si}N_A(2\phi_F)}}{C_{ox}} \cdot \frac{x_j}{2L} \left[\left(\sqrt{1 + \frac{2x_{dS}}{x_j}} - 1 \right) + \left(\sqrt{1 + \frac{2x_{dD}}{x_j}} - 1 \right) \right]$$

which gives the characteristic $\Delta V_T \propto x_j/L$ dependence.

> **Use this as intuition, not as a model.** Charge sharing is a one-dimensional story told about a two-dimensional problem. It predicts the right *trend* and it explains why shallow junctions help, which is why it survives in textbooks. But it contains no drain-voltage dependence, so it cannot explain DIBL at all, and it badly underestimates the roll-off below about 50 nm. At the dimensions this module is about, the scale-length picture of Section 1.1 is the one to reason with.

2 Symptoms: the gate loses control

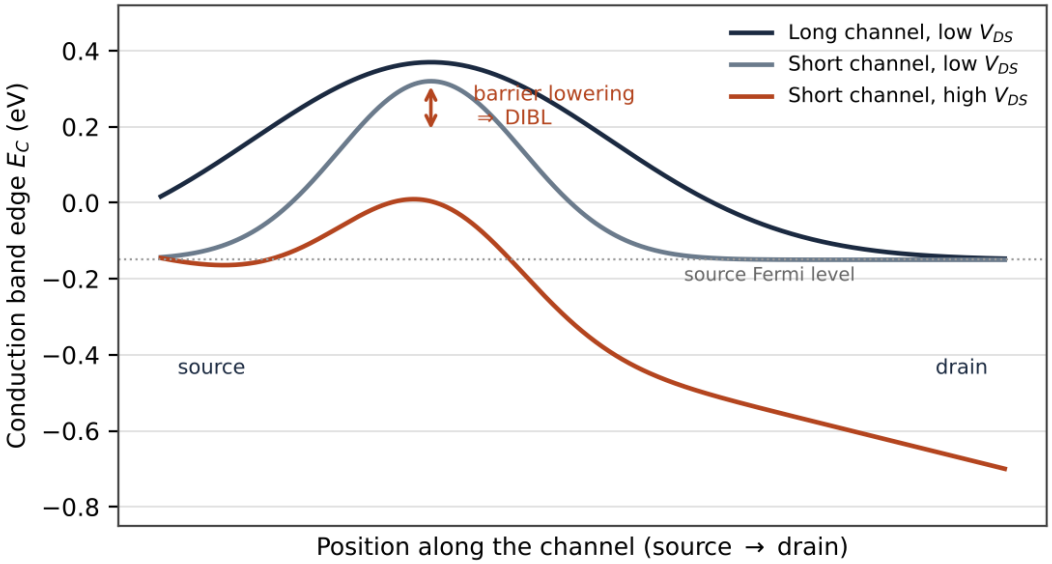


Figure 3.3. *Potential barrier between source and drain, which is the thing the gate is supposed to control. Long channel at low drain bias: a high, wide barrier set by the gate alone. Short channel: the barrier is lower and narrower even at low drain bias — that is V_T roll-off. Raise the drain voltage and the barrier falls further — that is DIBL. Punchthrough is the same figure with the barrier gone.*

Read that figure carefully, because the next four subsections are all in it.

2.1 Threshold-voltage roll-off

V_T falls as L falls. Mechanically: part of the depletion charge is supported by the junctions rather than the gate (Section 1.2), or equivalently, the source and drain raise the channel potential so less gate voltage is needed to invert (Section 1.1).

The consequence that matters is not that V_T is lower — you could compensate for that — but that V_T becomes a **steep function of** L. Gate length varies across a wafer by a nanometre or two from lithography alone. On the steep part of the roll-off curve in Figure 3.1, a 1 nm variation in L becomes tens of millivolts of V_T variation, and Section 5 shows what that does to a chip.

2.2 Drain-induced barrier lowering (DIBL)

Raise V_{DS} and the barrier falls further, so V_T falls with drain voltage. Quantified as

$$\text{DIBL} = -\frac{\Delta V_T}{\Delta V_{DS}} = \frac{V_T\big|_{V_{DS}=\text{low}} - V_T\big|_{V_{DS}=\text{high}}}{V_{DS,\text{high}} - V_{DS,\text{low}}} \quad [\text{mV/V}]$$

Typical values in a well-scaled modern device are 20–60 mV/V. Values above about 100 mV/V indicate poor electrostatic control.

DIBL is the symptom engineers actually measure, because it is a direct, single-number read-out of electrostatic integrity — it is the gap between the solid and dashed curves in Figure 3.1. It has two costs: the off-state current rises with drain bias, and output conductance in saturation rises, which destroys analogue gain.

> **Common pitfall.** Older texts say DIBL "can typically be eliminated by properly scaling the drain and source depths while increasing the substrate doping density." That was sound advice for a 0.25 μm bulk process. It is not true now: at $L \approx 15$ nm, junction and doping engineering alone cannot deliver the required λ, which is precisely why the industry changed the device geometry instead. Section 6 gives the modern answer.

2.3 Punchthrough

Push further and the source and drain depletion regions merge beneath the channel. The barrier in Figure 3.3 disappears, and current flows through the bulk, below the gate, where the gate has essentially no control over it. Drain current then rises steeply and almost uncontrollably with V_{DS}.

Punchthrough is not really a separate effect — it is the terminal case of the same loss of electrostatic control. It is suppressed by anything that raises the barrier deep in the body: higher body doping (classically), a retrograde or halo implant (Section 6.1), or simply removing the deep body altogether, which is what SOI and the multi-gate architectures do.

2.4 Subthreshold swing degradation

Document 2 established $SS = (1 + C_d/C_{ox})(kT/q)\ln 10$, with a floor of 59.5 mV/decade at 300 K. In a short device, the drain adds its own capacitive coupling to the channel, effectively

$$SS \approx \left(1 + \frac{C_d + C_{drain}}{C_{ox}}\right)\frac{kT}{q}\ln 10,$$

so the swing degrades beyond its long-channel value. 100–120 mV/decade is common in a badly scaled device.

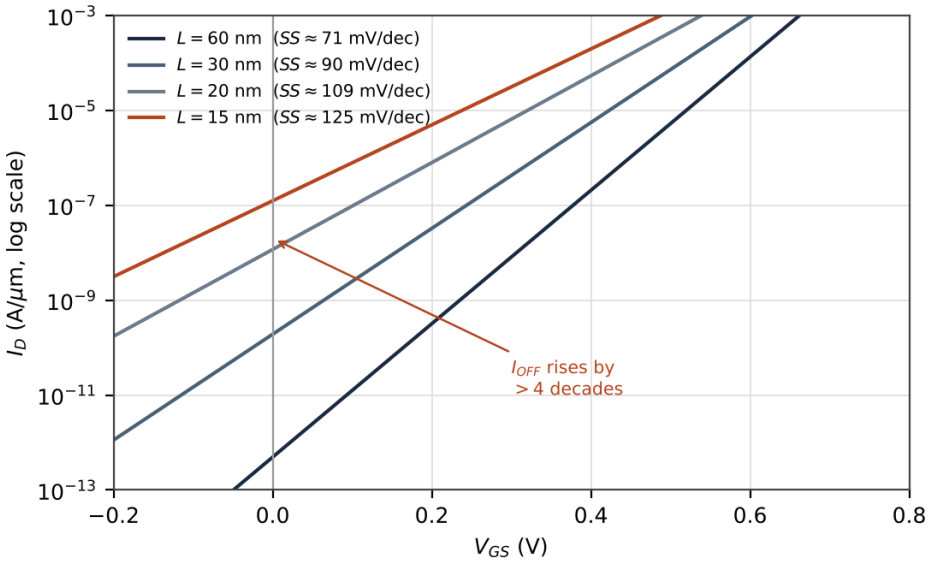


Figure 3.4. *Subthreshold characteristics as gate length is reduced, for a planar bulk device at $V_{DS} = 0.7$ V. Three degradations happen together and compound: V_T rolls off (the curves shift left), the swing degrades (they become less steep), and consequently the off-state current at $V_{GS} = 0$ rises by more than four orders of magnitude.*

Figure 3.4 is the single most important plot in this document, because it shows that **the three symptoms are not independent**. V_T roll-off shifts the curve left; DIBL shifts it further left as V_{DS} rises; swing degradation flattens it. All three push the same quantity — off-state current — in the same direction, and they multiply rather than add.

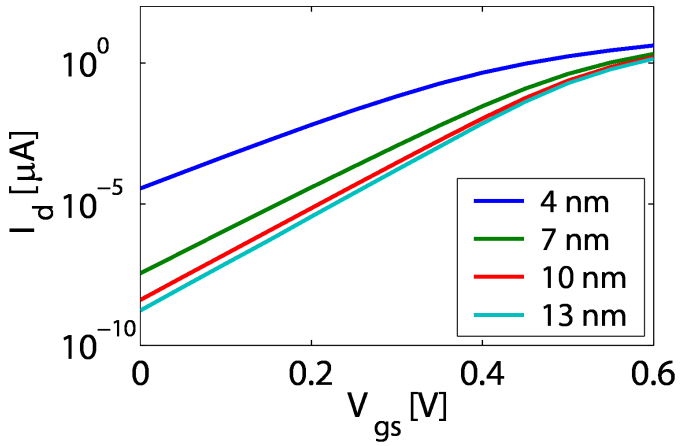


Figure 3.5. *Quantum-transport simulation of an ultra-scaled transistor as the gate length falls from 13 nm to 4 nm. The off-current rises by orders of magnitude and the swing flattens: the trends of Figure 3.4, computed from first principles. Reproduced from ETH Zürich lecture slides.*

2.5 Channel-length modulation

As V_{DS} rises past saturation, the pinch-off point moves toward the source and the *effective* channel length shortens, so current keeps rising instead of saturating:

$$I_D \approx I_{D,sat}(1 + \lambda_{CLM} V_{DS}).$$

Note that λ_{CLM} here is the circuit-level empirical parameter and has nothing to do with the natural length λ of Section 1.1 — an unfortunate collision of notation that this document flags rather than silently inherits.

CLM is mostly an analogue concern: it sets the intrinsic gain $g_m r_o$ of a transistor. In digital logic it is secondary to DIBL.

3 Symptoms: high fields

A 0.7 V supply across a 15 nm channel is an average field of about 4.7×10^5 V/cm, with local peaks several times higher near the drain. The following are consequences of operating there.

3.1 Velocity saturation

Document 2, Section 2.4 showed that drift velocity saturates near 10^7 cm/s. In a short channel the device is in saturation over most of its operating range, which changes the drive current law:

$$I_{D,sat} \approx W C_{ox} v_{sat} (V_{GS} - V_T)$$

Linear in overdrive, not quadratic — and independent of L. Three consequences worth stating plainly:

- Drive current stops improving when you shorten the channel. The classical reason for scaling L disappears.
- Transconductance $g_m \approx WC_{ox}v_{sat}$ becomes a geometric constant.
- The square-law model, and every circuit intuition built on it, stops applying.

This connects directly to the quasi-ballistic picture in Document 2, Section 4.1: at $L \approx \lambda_{mfp}$, drive current is set by source injection, not by transport along the channel.

3.2 Mobility degradation at the surface

The vertical field in the silicon at the interface of a modern device is of order 10^6 V/cm — several MV/cm across the dielectric itself — pressing carriers hard against the interface. Effective mobility falls to 200–400 $\mathrm{cm^2V^{-1}s^{-1}}$ against a bulk value above 1400.

The mechanisms — surface roughness scattering, remote Coulomb and remote phonon scattering from the high-k layer, and enhanced phonon scattering in the confined two-dimensional gas — are covered in Document 2, Section 4.3, including a common but incorrect explanation of them.

3.3 Impact ionisation and substrate current

A carrier that gains more than the bandgap energy between collisions can create an electron-hole pair on impact. In an n-channel device the generated electrons leave via the drain and the holes flow into the body, producing a **substrate current** that peaks at roughly $V_{GS} \approx V_{DS}/2$ — high enough field, still enough carriers in the channel.

The substrate current forward-biases the source–body junction. If the local potential rise reaches about 0.6 V, the parasitic source–body–drain npn bipolar turns on, and the positive feedback between avalanche multiplication and bipolar action causes breakdown well below the junction breakdown voltage.

This is largely a historical concern at modern supply voltages. At $V_{DD} = 0.7$ V, carriers cannot acquire the energy needed for impact ionisation — at least the 1.1 eV bandgap, and in practice nearer 1.5 eV because momentum must also be conserved — from the supply alone. It remains relevant in I/O devices, in high-voltage and analogue blocks on the same die, and — importantly — in SOI and FinFET devices where the body is floating, so even a small generated hole current has nowhere to go and the body potential floats up. The **floating body effect** is the modern descendant of this topic.

3.4 Hot carriers — and what actually limits reliability now

A "hot" carrier is one with energy far above the lattice thermal energy. If it exceeds the Si–dielectric barrier (≈ 3.1 eV for electrons, ≈ 4.8 eV for holes) it can be injected into the dielectric, where it is trapped, shifting V_T and degrading transconductance over the device lifetime.

> **What limits reliability now.** Hot-carrier injection was *the* reliability concern of the 5 V era (it is why the lightly doped drain was invented). At $V_{DD} < 1$ V a channel carrier cannot reach 3.1 eV, and classical injection over the barrier is largely suppressed. Hot-carrier degradation has not gone away, but its dominant mechanism has changed: interface bonds are now broken mainly by the cumulative energy of many moderately energetic carriers, rather than by single carriers crossing the barrier, and self-heating

in FinFET and nanosheet structures aggravates it. It now shares the reliability budget with:

- **BTI** (NBTI for pMOS, PBTI for nMOS) — bias-temperature instability, a V_T shift under gate bias at temperature, usually the leading wear-out mechanism in current technologies;
- **TDDB** — time-dependent dielectric breakdown, the slow degradation of the gate dielectric under field;
- **electromigration** in the interconnect, which is not a transistor effect at all but is often the actual lifetime limiter in a product.

4 Symptoms: leakage

Every effect in Section 2 ultimately shows up as current flowing when the device should be off. Because a modern die holds of order 10^{10} transistors, the off-state current of a single device — a few nanoamps or less — aggregates into watts.

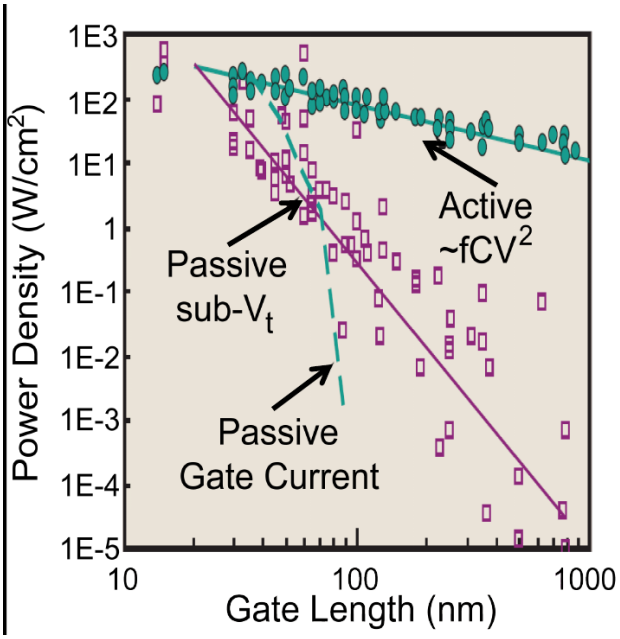


Figure 3.6. *Active (switching) and passive (leakage) power density against gate length. As the gate shrinks, sub-threshold and gate-leakage power climb towards the active power — the crossover behind the end of Dennard scaling. Source: W. Haensch et al., IBM J. Res. Dev. 50, 339 (2006); reproduced from ETH Zürich lecture slides.*

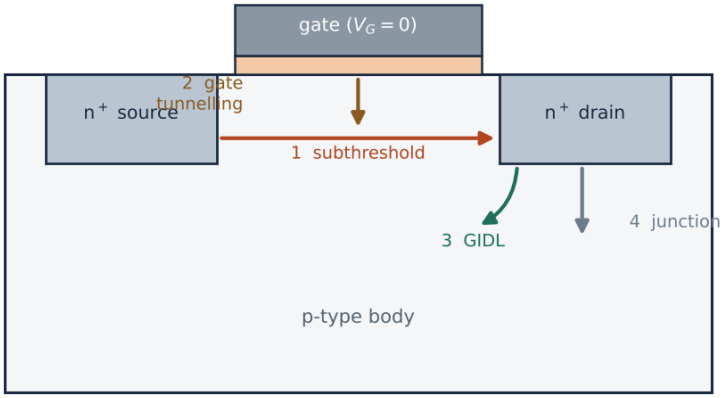


Figure 3.7. *The four leakage paths in an off-state n-channel MOSFET. Only the first is "short-channel" in origin, but all four are made worse by scaling.*

1. Subthreshold conduction. The dominant component, governed by V_T and SS, and therefore made worse by roll-off, DIBL and swing degradation together (Figure 3.4). Document 2, Section 3.5 derives it.

2. Gate direct tunnelling. Carriers tunnel through the dielectric. The current depends exponentially on physical thickness, which is why SiO_2 became untenable below about 1.2 nm and why high-k dielectrics were adopted (Document 1, Section 5.2) — the same capacitance at four to five times the physical thickness.

3. Gate-induced drain leakage (GIDL). With the gate at 0 V and the drain at V_{DD}, the gate-to-drain overlap region sees a large field across the dielectric. The drain surface is driven into deep depletion, bands bend steeply enough for **band-to-band tunnelling**, and carriers are generated directly in the drain. GIDL rises with drain doping and with thinner dielectrics, and it sets a floor on off-current that subthreshold engineering cannot get below.

GIDL is often omitted from introductory treatments. It matters because in low-power technologies it is frequently the limiting leakage component.

4. Junction leakage. Reverse-biased drain–body junction leakage, including a band-to-band component at the high doping levels now used.

The engineering consequence

$$I_{OFF} \times N_{transistors} \times V_{DD} = P_{static}$$

This product is why the entire preceding discussion matters commercially, and it closes the loop to Document 1, Section 4: static power became comparable to switching power in the early 2000s, V_T scaling stopped, and Dennard scaling ended. The chain runs from two-dimensional electrostatics in a single device to the architecture of every computer built since.

5 Symptoms: variability

Everything so far has described the *average* device. At nanoscale dimensions the spread between nominally identical devices becomes comparable to the design margin.

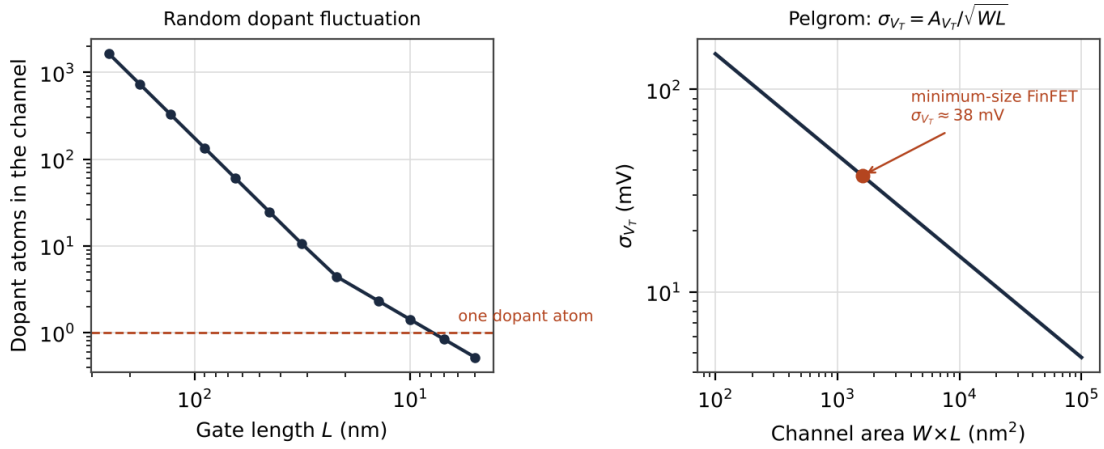


Figure 3.8. *Left: the number of dopant atoms in the channel volume as gate length is reduced — an estimate using doping that rises as L falls, as classical scaling requires, and a square channel ($W = L$). Below roughly 45 nm gate length there are only tens of atoms, and the discreteness of matter becomes a circuit-design problem. Right: Pelgrom scaling, $\sigma_{V_T} = A_{V_T}/\sqrt{WL}$, drawn for an illustrative $A_{V_T} = 1.5$ mV· μm. The marked point is a minimum-size FinFET (effective width ≈ 100 nm, $L \approx 16$ nm): σ_{V_T} of roughly 40 mV, so a 5σ outlier is displaced by a large fraction of V_T itself.*

Random dopant fluctuation (RDF). A channel of 15 × 15 × 20 nm doped at 10^{18} cm^{-3} contains about **five dopant atoms**. Not five million; five. Their number is Poisson-distributed and their positions are random, so V_T is a random variable:

$$\sigma_{V_T} \approx \frac{q}{C_{ox}}\sqrt{\frac{N_A\, W_{dm}}{3\,WL}} \quad \propto \quad \frac{1}{\sqrt{WL}}$$

This is the deepest reason the industry abandoned doped channels. In a fully depleted SOI, FinFET or nanosheet device the channel is nominally **undoped**, V_T is set by the gate metal work function instead, and RDF largely disappears. That is a motivation for the architecture change at least as important as electrostatics, and it is rarely taught alongside it.

Line-edge and line-width roughness (LER/LWR). Lithographic edges are rough at the 1–2 nm level. On a 12 nm gate that is over 10% of the critical dimension, and because V_T depends steeply on L (Section 2.1), edge roughness converts directly into threshold spread.

Metal gate granularity. A metal gate is polycrystalline, and different grain orientations have different work functions. With only a few grains over a nanoscale gate, the effective work function — and hence V_T — varies device to device.

Why it matters. SRAM is the worst case: six minimum-size transistors that must remain balanced. SRAM stability, not logic speed, frequently sets the minimum operating voltage of a whole chip. The consequence is that device design has become a **statistical** discipline. You do not design the nominal device; you design the distribution's tail, at 5σ or beyond, because with 10^{10} devices on a die, a one-in-a-million outlier occurs ten thousand times.

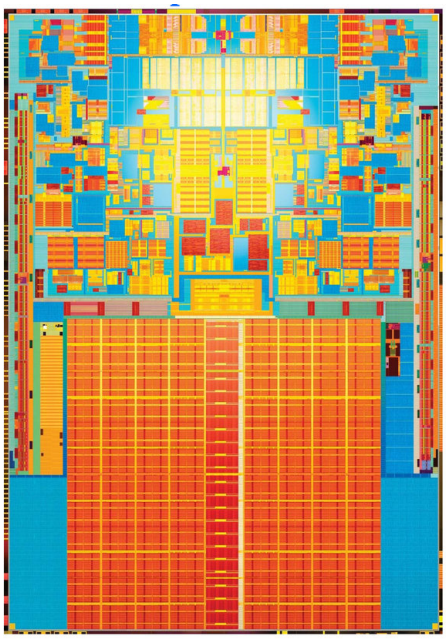


Figure 3.9. *Die photograph of a 45 nm-generation Intel processor (2008). The large, regular arrays are on-chip SRAM cache; the irregular regions are logic. SRAM is where variability bites first. Image: Intel; reproduced from W. A. Doolittle, Georgia Tech lecture slides (2011).*

6 One family of solutions

Return to Section 1.1. Every symptom in Sections 2–5 traces to the drain reaching into the channel, and the design rule is $L \gtrsim 5\lambda$ with

$$\lambda = \sqrt{\frac{\varepsilon_{Si}}{\varepsilon_{ox}} t_{Si} t_{ox}}\,.$$

There are only four things you can do to that equation, and the history of the last twenty-five years consists of doing each in turn.

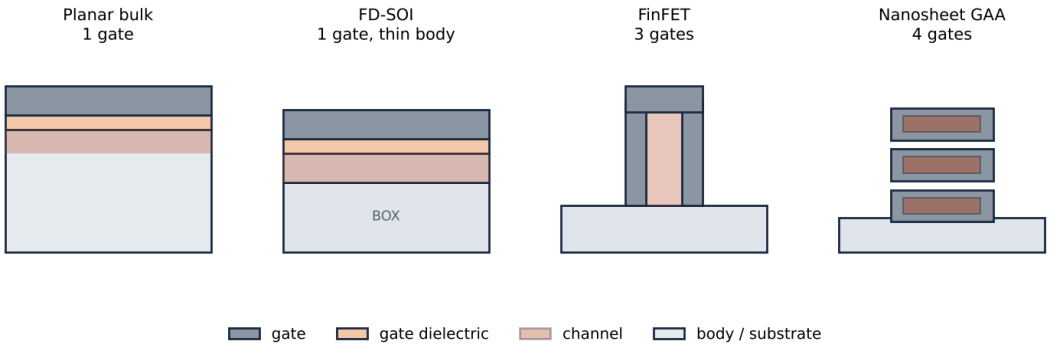


Figure 3.10. *The architectural sequence, in cross-section perpendicular to current flow. Each step reduces the volume of semiconductor the drain field can reach through, and increases the number of sides from which the gate screens it.*

6.1 Lever 1: reduce t_{ox} — better gate coupling

Thin the dielectric, then — once direct tunnelling forbids thinning further — raise its permittivity so the *electrical* thickness falls while the physical thickness rises. This is the high-k/metal gate transition of 2007 (Document 1, Section 5.2). It is a square-root lever and

it has been exhausted: EOT is now around 0.8–1 nm, with a further 0.3–0.4 nm of unavoidable quantum-mechanical penalty (Document 2, Section 4.2).

6.2 Lever 2: reduce x_j and engineer the doping — the classical fixes

Shallow junctions reduce the charge-sharing geometry directly. **Halo or pocket implants** place locally higher doping next to the source and drain to raise the barrier where the drain would otherwise lower it.

Halos work, and they carried bulk planar CMOS through its last decade, to the 28/20 nm class. They also carry three costs that eventually made them untenable: they increase junction field and therefore GIDL; they produce the **reverse short-channel effect**, a non-monotonic $V_T(L)$ that is awkward for design; and above all they require heavy local doping, which maximises RDF (Section 5) exactly when variability was becoming critical. They are not used in fully depleted devices.

6.3 Lever 3: reduce t_{Si} — thin the body

If there is less silicon, there is less volume for the drain field to reach through. **Fully depleted SOI** places a thin silicon film over a buried oxide, cutting off the deep leakage paths entirely — no punchthrough beneath the channel, because there is no "beneath". The body is undoped, so RDF largely disappears, and V_T is set by the gate work function. FD-SOI remains in volume production at the 28 and 22 nm classes for low-power and RF products, where its back-gate bias control is valuable.

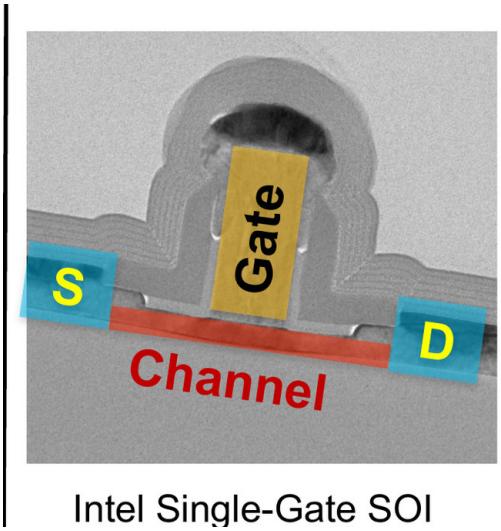


Figure 3.11. *Intel's experimental single-gate SOI transistor: a thin silicon channel on buried oxide, so there is no deep body for the drain field to reach through. Source: R. Chau et al., Device Research Conference 2003; reproduced from ETH Zürich lecture slides.*

6.4 Lever 4: add gates — the prefactor

Screening the drain from more than one side divides λ^2 by the number of effective gates. This is the FinFET ($\approx$ 3 gates) and the gate-all-around nanosheet ($\approx$ 4). It is the strongest lever available and it is the reason for the two architecture changes in Document 1's timeline.

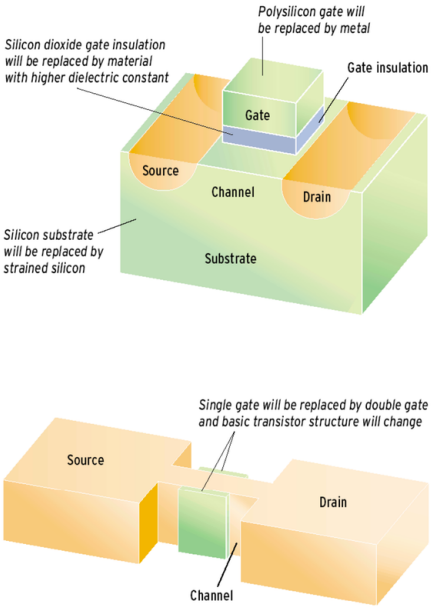


Figure 3.12. *A prediction from 2002. The upper drawing marks the planned replacements in a planar transistor — high-k for SiO_2, metal for polysilicon, strained silicon for the substrate — and the lower one the move from a single gate to a double gate. All four have since happened. Source: IEEE Spectrum, October 2002; reproduced from W. A. Doolittle, Georgia Tech lecture slides (2011), after O. Brandt.*

As the table in Section 1.1 showed, these levers together move the minimum usable gate length from about 39 nm (planar bulk) to about 15 nm (FinFET) to about 10 nm (GAA). **That table is the roadmap of this course**, and Lectures 2–4 are what comes after the levers in this equation run out.

6.5 What happens when they do run out

Three directions:

- **Keep improving electrostatics geometrically** — CFET stacking (Document 1, Section 5.6). This buys density, not a better λ.
- **Change the channel material** for lower effective mass and higher injection velocity: III–V compound semiconductors and carbon nanotubes (*Lecture 4*), and two-dimensional materials such as MoS_2.
- **Change the injection mechanism** to escape the 60 mV/decade floor altogether, since that floor — not electrostatics — is now the binding constraint on V_{DD}. Tunnel FETs use band-to-band tunnelling in place of thermionic emission; Lecture 3 develops the tunnelling physics they rely on.

Note the logic of that last one. Sections 1–6 have been about restoring the gate's control of the barrier. Once the gate has full control, SS reaches 60 mV/dec and stops — a limit that is thermodynamic, not electrostatic. Perfect electrostatics is not enough. **That is why the module continues past CMOS**, and it is the thread connecting Lecture 1 to everything that follows.

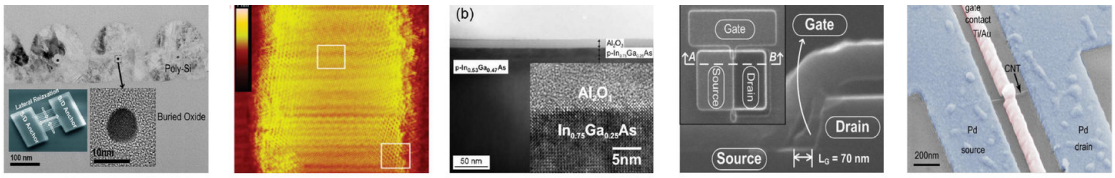


Figure 3.13. *Post-silicon candidates as presented around 2009: nanowire, graphene, III–V ultra-thin body, band-to-band tunnelling FET and carbon nanotube. The nanowire became today's nanosheet; the others remain research. Sources: P. Hashemi et al., IEEE EDL 30, 401 (2009); L. Tapasztó et al., Nat. Nanotechnol. 3, 397 (2008); Y. Q. Wu et al., IEEE EDL 30, 700 (2009); W. Y. Choi et al., IEEE EDL 28, 743 (2007); S. Guha, IBM Research; reproduced from ETH Zürich lecture slides.*

7 In-class animations

Eleven interactive animations go with this lecture. Each is a single web page with the animation, things to try in class, check questions with answers, and a statement of the model behind it. The full set is at faizverse.github.io/nanoelectronic-devices/L1; the last column gives each page's file name in that folder.

#	Animation	Section	Page
1	MOSFET basics	**Document 2, Section 3**	mosfet-basics.html
2	Natural length and threshold roll-off	Sections 1.1 and 2.1	vt-rolloff.html
3	Drain-induced barrier lowering	Section 2.2	dibl.html
4	Punchthrough	Section 2.3	punchthrough.html
5	Subthreshold swing degradation	Section 2.4	subthreshold-swing.html
6	Channel-length modulation	Section 2.5	channel-length-modulation.html
7	Carrier velocity saturation	Section 3.1	velocity-saturation.html
8	Hot carriers and impact ionisation	Sections 3.3 and 3.4	hot-carriers.html
9	Oxide breakdown (TDDB)	Section 3.4	oxide-breakdown.html
10	Gate-induced drain leakage (GIDL)	Section 4	gidl.html
11	Variability: random dopants	Section 5	variability.html

Animation 8 shows the classical lucky-electron injection mechanism, and why it disappears at a 0.7 V supply; read it alongside the box in Section 3.4. Each page also works offline once saved, which is useful in rooms without a reliable network.

8 Summary

The one idea. The gate must control the source-to-drain barrier. Short channels let the drain compete. Everything else is consequence.

The one equation.

$$\lambda = \sqrt{\frac{\varepsilon_{Si}}{\varepsilon_{ox}} t_{Si} t_{ox}}, \qquad L \gtrsim 5\lambda$$

The symptoms, grouped by cause:

Group	Effects
Loss of gate control (Section 2)	V_T roll-off, DIBL, punchthrough, SS degradation, CLM
High fields (Section 3)	velocity saturation, mobility degradation, impact ionisation, hot carriers
Leakage (Section 4)	subthreshold, gate tunnelling, GIDL, junction
Variability (Section 5)	RDF, LER/LWR, metal gate granularity

The solutions, as four levers on one equation (Section 6): thinner effective dielectric; shallower junctions and halo doping; thinner body; more gates. Together they take L_{min} from 39 nm to about 10 nm.

The limit beyond them. Perfect electrostatics still leaves SS at 60 mV/decade, because that floor is thermodynamic. Escaping it requires a different injection mechanism or a different channel material — which is Lectures 2, 3 and 4.

Sources and further reading

Scale length and electrostatic integrity:

- R.-H. Yan, A. Ourmazd and K. F. Lee, "Scaling the Si MOSFET: from bulk to SOI to bulk", *IEEE Trans. Electron Devices*, vol. 39, no. 7, pp. 1704–1710, 1992. *The natural-length paper. Short; read it.*
- C. P. Auth and J. D. Plummer, "Scaling theory for cylindrical, fully-depleted, surrounding-gate MOSFETs", *IEEE Electron Device Letters*, vol. 18, no. 2, pp. 74–76, 1997. *Extends the result to gate-all-around.*
- J.-P. Colinge, "Multiple-gate SOI MOSFETs", *Solid-State Electronics*, vol. 48, no. 6, pp. 897–905, 2004. *The effective-number-of-gates generalisation used for the double-gate and GAA forms in Section 1.1.*
- J.-P. Colinge (ed.), *FinFETs and Other Multi-Gate Transistors*, Springer, 2008.
- IRDS, *International Roadmap for Devices and Systems, 2022 Edition: More Moore*, IEEE, 2022. https://irds.ieee.org/images/files/pdf/2022/2022IRDS_MM.pdf

Short-channel effects and scaling generally:

- Y. Taur and T. H. Ning, *Fundamentals of Modern VLSI Devices*, 3rd ed., Cambridge University Press, 2021, Chapters 3 and 7.
- Y. Taur, "CMOS design near the limit of scaling", *IBM Journal of Research and Development*, vol. 46, no. 2/3, pp. 213–222, 2002.

Leakage:

- K. Roy, S. Mukhopadhyay and H. Mahmoodi-Meimand, "Leakage current mechanisms and leakage reduction techniques in deep-submicrometer CMOS circuits", *Proceedings of the IEEE*, vol. 91, no. 2, pp. 305–327, 2003. *The standard reference on the four leakage paths of Section 4, including GIDL.*

Variability:

- A. Asenov, “Random dopant induced threshold voltage lowering and fluctuations in sub-0.1 μm MOSFETs”, *IEEE Trans. Electron Devices*, vol. 45, no. 12, pp. 2505–2513, 1998.
- M. J. M. Pelgrom, A. C. J. Duinmaijer and A. P. G. Welbers, “Matching properties of MOS transistors”, *IEEE J. Solid-State Circuits*, vol. 24, no. 5, pp. 1433–1439, 1989.

Figure credits. Figures 3.1, 3.4 and 3.8 are computed from the models and parameters stated in their captions; they illustrate scaling behaviour rather than measure any particular technology. Figures 3.2, 3.3, 3.7 and 3.10 are schematics drawn for this document; Figure 3.2 follows the classical charge-sharing construction. The other figures are reproduced from earlier course material for this module — ETH Zürich lecture slides and W. A. Doolittle’s Georgia Tech slides (2011) — with the original source, where known, given in each caption. Copyright in reproduced figures remains with the original owners.